

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411013
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Takahashi; Yusuke et al.

Mobile body, method of controlling mobile body, and program

Abstract

A mobile body controller according to the present disclosure includes circuitry configured to recognize an environment surrounding a mobile body to be controlled, and change parameters used for self-position estimation by the mobile body based on the recognized environment.

Inventors:	Takahashi; Yusuke (Kanagawa, JP), Watanabe; Ryo (Tokyo, JP), Saito; Yukihiro (Tokyo, JP), Nabeta; Masaomi (Tokyo, JP)
Applicant:	SONY GROUP CORPORATION (Tokyo, JP)
Family ID:	68965963
Assignee:	SONY GROUP CORPORATION (Tokyo, JP)
Appl. No.:	17/299366
Filed (or PCT Filed):	December 03, 2019
PCT No.:	PCT/JP2019/047290
PCT Pub. No.:	WO2020/121900
PCT Pub. Date:	June 18, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220019224 A1	Jan. 20, 2022

Foreign Application Priority Data

JP	2018-231033	Dec. 10, 2018
----	-------------	---------------

Publication Classification

Int. Cl.: **G01C22/00** (20060101); **B60W60/00** (20200101); **G01C21/16** (20060101); **G01C21/28** (20060101); **G05D1/00** (20060101)

U.S. Cl.:

CPC **G01C21/28** (20130101); **B60W60/001** (20200201); **G01C21/1656** (20200801); **G05D1/0212** (20130101); **G05D1/024** (20130101); **G05D1/0272** (20130101); **G05D1/0274** (20130101); **G05D1/0278** (20130101);

Field of Classification Search

CPC: G05D (1/0297); G05D (1/0278); G05D (1/0272); G05D (1/0274); G05D (1/024); G05D (1/0212); G05D (1/0253); G06Q (10/063114); G06Q (10/06316); G06Q (10/0835); G01C (21/28); G01C (21/1656); B60W (60/001)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11199853	12/2020	Afrouzi	N/A	B25J 13/006
2006/0129276	12/2005	Watabe	701/23	G05D 1/024
2017/0167881	12/2016	Rander	N/A	G05D 1/0274
2018/0003516	12/2017	Khasis	N/A	G08G 1/207
2018/0204338	12/2017	Narang et al.	N/A	N/A
2020/0097006	12/2019	Liu	N/A	G05B 19/4155

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H06-348971	12/1993	JP	N/A
2015170127	12/2014	JP	N/A
2017109781	12/2016	JP	N/A
2018206004	12/2017	JP	N/A
WO-2016059930	12/2015	WO	N/A

OTHER PUBLICATIONS

Meimei Gao et al., Control Strategy Selection for Autonomous Vehicles in a Dynamic Environment, 2005 IEEE International Conference on Systems, Man and Cybernetics, Oct. 2005, pp. 1-6, IEEE. cited by applicant

Primary Examiner: Alharbi; Adam M

Attorney, Agent or Firm: Paratus Law Group, PLLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2019/047290 filed on Dec. 3, 2019 under 35 U.S.C. § 371, which claims the benefit of Japanese Priority Patent Application JP 2018-231033 filed on Dec. 10, 2018, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a mobile body, a method of controlling a mobile body, and a program.

BACKGROUND ART

(3) In related art, PTL 1 below, for example, describes estimating a position and an attitude of a mobile body using map data of a traveling environment and geometric data which is a sensing result of a distance sensor unit, and switching a traveling mode between a state in which the position and attitude of the mobile body are fixed uniquely and a state in which the position and attitude of the mobile body are not fixed uniquely.

CITATION LIST

Patent Literature

(4) PTL 1: JP 6348971 B2

SUMMARY

Technical Problem

(5) However, the technique described in the patent literature above assumes switching the traveling mode in accordance with the environment, but allows the state in which the position and attitude of the mobile body are not fixed uniquely, and thus involves a probability that the mobile body itself cannot recognize its own position (loses its own position).

(6) Thus, it has been requested to increase the localization accuracy without being affected by changes in traveling environment.

Solution to Problem

(7) According to an aspect of the present disclosure, there is provided a mobile body controller including: circuitry configured to recognize an environment surrounding a mobile body to be controlled, and change parameters used for self-position estimation by the mobile body based on the recognized environment.

(8) Further, according to another aspect of the present disclosure, there is provided a mobile body including: one or more sensors; and a mobile body controller, the mobile body controller including circuitry configured to recognize an environment surrounding the mobile body controlled by the mobile body controller, and change parameters used for self-position estimation by the mobile body based on the recognized environment.

(9) Further, according to another aspect of the present disclosure, there is provided a mobile body control method including: recognizing an environment surrounding a mobile body to be controlled; and changing parameters used for self-position estimation by the mobile body based on the recognized environment.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing a configuration of a mobile body controller according to an embodiment of the present disclosure.

(2) FIG. 2A is a schematic view showing a route along which a mobile body travels.

(3) FIG. 2B is a schematic view showing a route along which a mobile body travels.

(4) FIG. 3A is a schematic view showing target positions by dots when a mobile body travels along the paths shown in FIGS. 2A and 2B.

- (5) FIG. 3B is a schematic view showing target positions by dots when a mobile body travels along the paths shown in FIGS. 2A and 2B.
- (6) FIG. 4 is a schematic view showing a configuration of a system in a case of embedding environmental information in an advance map.
- (7) FIG. 5 is a flowchart showing processing in the case of embedding environmental information in the advance map.
- (8) FIG. 6 is a schematic view showing a configuration of a system in a case where a mobile body itself recognizes an environment.
- (9) FIG. 7 is a schematic view showing an example of localization profile.
- (10) FIG. 8 is a flowchart showing processing in a case where a mobile body itself recognizes an environment.
- (11) FIG. 9 is a schematic view showing a configuration of a system in a case of performing path planning in accordance with a map having parameters embedded and a sensor situation.
- (12) FIG. 10 is a flowchart showing processing in the case of performing path planning in accordance with the map having parameters embedded and the sensor situation.

DESCRIPTION OF EMBODIMENTS

- (13) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the appended drawings. Note that, in this specification and the appended drawings, structural elements that have substantially the same function and structure are denoted with the same reference numerals, and repeated explanation of these structural elements is omitted.
- (14) Note that description will be provided in the following order.
- (15) 1. Overview of present disclosure
- (16) 2. Method of localization
 - (17) 2.1. Dead reckoning
 - (18) 2.2. SLAM
 - (19) 2.3. Visual odometry
 - (20) 2.4. GPS
 - (21) 2.5. Wheel odometry
 - (22) 2.6. IMU
- (23) 3. Localization by extended Kalman filter
- (24) 4. Examples of parameter in accordance with environment
 - (25) 4.1. Environment in which nothing exists in surroundings
 - (26) 4.2. Sandy place
 - (27) 4.3. Outdoor environment surrounded by buildings
 - (28) 4.4. Corridor environment adjacent to glass walls
 - (29) 4.5. Office environment with many repeated patterns
- (30) 5. Method of changing parameter
 - (31) 5.1. Example of embedding environmental information in advance map
 - (32) 5.2. Case in which mobile body itself recognizes environment
 - (33) 5.3. Example of embedding environmental information in advance map and performing path planning in accordance with sensor situation
- (34) 1. Overview of Present Disclosure
- (35) When making an autonomous movement, a mobile body such as a robot generates a surrounding environment map in advance, estimates its own position on the map on the basis of sensing information, and travels toward a destination. On this occasion, an amount of control is calculated with reference to the estimated position of the mobile body and a path directed to the destination. Therefore, the localization accuracy is an important factor that determines whether or not autonomous traveling can be performed.
- (36) Various techniques exist as localization techniques, which are great research themes in robotics. The respective localization techniques are good at and poor at different environments in

terms of properties of used sensors and algorithms. In addition, since many setting parameters exist in the respective techniques, tuning in accordance with the environment is necessary in some cases. (37) In addition, in a case of assuming improving robustness of localization by sensor fusion, it is desirable to also tune a parameter that determines the weight of sensors to be utilized for fusion in accordance with the environment. Consequently, in order to move autonomously across a plurality of traveling environments having different properties, it is desirable to change the localization technique and parameter appropriately in accordance with the respective environments.

(38) In an embodiment, concerning localization of a mobile body such as a robot, setting parameters for localization tuned to environments different in properties are changed dynamically in accordance with an environment in which the mobile body is traveling in order to achieve highly robust localization. Accordingly, a setting parameter for localization, used sensors, a localization algorithm, and processing settings are changed. Note that a mobile body that travels with wheels on the ground is shown as an example in an embodiment, but the mobile body may be a flying drone or the like.

(39) 2. Method of Localization

(40) As a method of localization, dead reckoning, simultaneous localization and mapping (SLAM), visual odometry (VO), global positioning system (GPS), wheel odometry, and inertial measurement unit (IMU) below will be shown as examples. Note that the method of localization is not limited to them, but another method may be used. Hereinafter, characteristics of the respective methods will be described.

(41) 2.1. Dead Reckoning

(42) Dead reckoning is a technique for estimating a relative position of a mobile body utilizing an internal sensor. By integrating and accumulating the speed or angular velocity obtained from an inertial sensor or a wheel odometry, an amount of movement of the mobile body itself is obtained from a reference point (original point). Although errors may be increased with time since the amount of movement is accumulated, a continuous and precise position of the mobile body can be estimated if only for a relatively short while.

(43) 2.2. SLAM

(44) SLAM is a technique for simultaneously estimating a position of a mobile body and a surrounding map using a laser scanner or a camera. The position of the mobile body is estimated while resolving a contradiction in view of how the surroundings of the mobile body are seen and the amount of movement of the mobile body, and an environment map is generated at the same time. In an environment without any characteristic feature, SLAM has a characteristic in which the localization accuracy decreases because correction with a map is difficult. In addition, SLAM has a property in which the localization accuracy decreases in the vicinity of glass, a mirror, or the like because noise is included in an observed value of a laser scanner.

(45) 2.3. Visual Odometry

(46) Visual odometry is a technique for estimating an amount of movement of a mobile body from an amount of changes with time of a feature amount within a camera image. The distance order may be different from actual measurement depending on camera calibration. A distance close to an actual measurement value can be obtained by utilizing stereoscopy, but the accuracy is relatively low in some cases. In addition, sight of a characteristic point may be lost in a light and dark environment, and the localization accuracy also decreases in some cases. In addition, it may also be difficult to extract the feature amount in a case where an image is blurred.

(47) 2.4. GPS

(48) GPS is a technique for receiving a signal emitted from a satellite to estimate a position of a receiver in a triangulation way. The absolute coordinates on the globe can be directly measured outdoors. In the vicinity of a building, a positioning error may occur because of an influence exerted by multipath such as a reflected wave or a diffracted wave. In addition, the positioning accuracy in an environment surrounded by high-rise buildings where a signal reaches incompletely

is low.

(49) 2.5. Wheel Odometry

(50) Wheel odometry is a technique for measuring and accumulating an amount of rotation of a tire to calculate an amount of movement of a mobile body, and is a kind of dead reckoning. Since an error occurs in the position of the mobile body when the tire slips, the localization accuracy may degrade on a slippery ground surface.

(51) 2.6. IMU

(52) IMU is called an inertial measurement unit, and detects an acceleration or an angular velocity. IMU is utilized when estimating an attitude of a mobile body and predicting a state in a current epoch from a state in a previous epoch. In the Kalman filter, IMU is often utilized when in processing called prediction update processing and time update processing.

(53) 3. Localization by Extended Kalman Filter

(54) In a case where a mobile body travels in an actual environment, various environments are assumed. Indoor examples include an environment in which there are many objects and a characteristic point is easy to extract, an environment in which there are many mirrors or glass, and a sensor such as a laser scanner is difficult to use, an environment without characteristics, such as a corridor, and the like.

(55) In addition, outdoor examples include an environment in which there are many buildings in the neighborhood, and noise is easily included in a GPS signal, an environment in which nothing exists in the surrounding, such as a square, an environment in which the traveling surface is sandy, and wheels are likely to slip, and the like.

(56) In such environments having different properties, it is desirable to perform localization with an optimized parameter using sensors of a combination suited to each of the environments. In an embodiment, a plurality of sensors and a plurality of localization techniques are fused by an extended Kalman filter.

(57) FIG. 1 is a schematic view showing a configuration of a system **1000** of a mobile body according to an embodiment of the present disclosure. As shown in FIG. 1, this system **1000** has an IMU **100**, a wheel odometry **110**, a visual odometry **120**, an SLAM **130**, and a GPS **140** as functional blocks that perform localization. In addition, the system **1000** has a magnetic sensor **150**, a camera **200**, and a laser scanner **210** as functional blocks (sensors) that perform localization. The camera **200** is utilized by the visual odometry **120**, and the laser scanner **210** is utilized by the SLAM **130**. In addition, the magnetic sensor **150** detects an orientation. In addition, the system **1000** has an extended Kalman filter **300** that fuses these localization techniques. Note that the respective structural elements of the system **1000** shown in FIG. 1 can be configured from hardware such as sensors or circuits, or a central processing unit such as CPU, and a program for causing this to function. The same applies to FIGS. 4, 6, and 9 which will be described later.

(58) As an example, in localization through use of the extended Kalman filter **300**, a position of the mobile body in the current time sample is predicted from a previous time sample by dead reckoning by means of the IMU **100** and the wheel odometry **110**, and a predicted value is corrected using observed values acquired in the current time sample to estimate the position of the mobile body. Prediction of the position of the mobile body in the current time sample from the previous time sample is performed in a time update block **310**. In addition, correction of the predicted value through use of the observed values acquired in the current time sample is performed in an observation update block **320**. Note that the observed values are mainly obtained from the visual odometry **120**, the SLAM **130**, the GPS **140**, and the magnetic sensor **150**, and are fused on the basis of a covariance value.

(59) In an embodiment, in order to obtain the predicted value, internal sensors such as the IMU **100** and the wheel odometry **110** are utilized. The extended Kalman filter **300** can integrate various sensor values and localization techniques by appropriately defining a model that represents a relation between observed values and an estimated value. However, covariance of the respective

observed values is appropriately set as a setting parameter that determines the reliability of the observed values. As the covariance value is smaller, sensor values and localization with higher reliability are achieved, and as the covariance value is larger, sensor values and localization with lower reliability are achieved. An embodiment shows an example of dynamically changing the covariance value of the extended Kalman filter **300** in accordance with a traveling environment of the mobile body.

(60) 4. Examples of Parameter in Accordance with Environment

(61) FIGS. 2A and 2B are schematic views showing routes along which a mobile body **10** travels. As shown in FIGS. 2A and 2B, the mobile body **10** first passes through a square **30** to a sandy place **40**, follows a road **50** surrounded by buildings **20** nearby, and enters into a building **40** including an office.

(62) FIG. 2B shows the inside of the building **40**. The mobile body **10** follows a corridor **54** with glass walls **52** existing laterally, and passes between desks **60** to reach a target point **90** (a seat **70**).

(63) As shown in FIG. 1, sensors used for localization of an autonomous mobile body shall be the IMU **100**, the wheel odometry **110**, the laser scanner **210**, the GPS **140**, the camera **200**, and the magnetic sensor **150**, and sensor fusion shall be performed in the extended Kalman filter **300**. However, the laser scanner **210** shall utilize the SLAM **130** that performs ICP matching, and the camera **200** shall utilize the visual odometry **120**.

(64) Hereinafter, examples of parameter in accordance with the environment will be described in a case where the mobile body **10** travels along the paths shown in FIGS. 2A and 2B.

(65) 4.1. Environment in which Nothing Exists in Surroundings

(66) In a case where the mobile body **10** travels outdoors, and travels in an environment in which there is no feature in the surroundings, the SLAM **130** is not useful because no outstanding landmark exists although there is no particular error factor. In such an environment, a positioning signal of the GPS **140** is also used mainly to perform localization. That is, when traveling through the square **30** in which no feature exists in the surroundings, it is less likely to be affected by multipath that will be an error factor of the GPS **140**, so that the reliability of the GPS **140** is increased. Consequently, in the extended Kalman filter **300**, it is desirable to make the covariance value of GPS smaller and make the covariance value of the SLAM **130** larger.

(67) 4.2. Sandy Place

(68) In a case where the mobile body **10** travels the sandy place **40** which is an outdoor environment but has a slippery ground surface, the wheel odometry **110** is likely to produce an error, so that the reliability of dead reckoning performed by the wheel odometry **110** decreases. Thus, dead reckoning performed by the IMU **100** is utilized without utilizing the wheel odometry **110**. In addition, since no feature exists in the surroundings at the sandy place **40** and the GPS **140** can be utilized similarly to the square **30**, the covariance of the GPS **140** is continuously made smaller.

(69) 4.3. Outdoor Environment Surrounded by Buildings

(70) Since a multipath error is included in a positioning solution of the GPS **140** in an outdoor environment such as the road **50** surrounded by the buildings **20**, it is desirable to make the covariance value of the GPS **140** larger. In addition, since it is difficult to assume that a wheel of the mobile body **10** slips if the road **50** is paved, the covariance of the wheel odometry **110** is made smaller.

(71) 4.4. Corridor Environment Adjacent to Glass Walls

(72) As shown in FIG. 2B, in a case where the mobile body **10** travels the corridor **54** adjacent to the glass walls **52**, the GPS **140** is not used since it is indoor traveling. In addition, since the glass walls **52** transmit or reflect laser light, an error occurs in positioning information obtained by the laser scanner **210**. Consequently, in such an environment with much glass, it is desirable to decrease the reliability of the SLAM **130** through use of the laser scanner **210**. Thus, in the extended Kalman filter **300**, the covariance of the position of the mobile body obtained from the

SLAM **130** is set to be large.

(73) 4.5. Office Environment with Many Repeated Patterns

(74) As shown in FIG. 2B, inside the building **40**, there are many objects, and regular repeated patterns such as the desks **60** exist. Since a characteristic point is easy to extract in such an environment, it is suitable to utilize the visual odometry **120** and the SLAM **130**. On the other hand, in a case where processing of matching with a map is included in the SLAM **130**, it is an environment in which mismatching is likely to occur because regular repeated patterns exist. Consequently, it is effective to decrease a mismatching occurrence rate by stopping map matching processing in the SLAM **130** or slowing the cycle of the map matching processing, rather than setting the covariance of the extended Kalman filter **300**.

(75) Setting parameters in accordance with the environment in which localization is performed as in the above examples is suitable for robust localization. In addition, as in the example of an office environment with many repeated patterns, it is desirable not only to change the covariance matrix of the extended Kalman filter **300**, but also to change a parameter in each of the localization techniques such as the SLAM **130**.

(76) 5. Method of Changing Parameter

(77) The examples of setting a parameter for localization in accordance with the environment have been described above. Hereinafter, techniques for actually changing a parameter for localization in accordance with a traveling environment will be described. In order to change the parameter for localization, the mobile body **10** needs to be supplied with information about a traveling environment in advance or the mobile body **10** itself needs to recognize the traveling environment.

(78) Here, a technique for supplying a parameter for localization together with positional information about a destination that the mobile body **10** will utilize at the time of navigation, and a technique for the mobile body to recognize an external environment using an identification device for changing a parameter profile for localization will be described.

(79) 5.1. Example of Embedding Environmental Information in Advance Map

(80) FIGS. 3A and 3B are schematic views showing target positions **80** by dots when the mobile body **10** travels along the paths shown in FIGS. 2A and 2B. When the mobile body **10** moves autonomously, the mobile body **10** is supplied with the target point **90** in advance. The mobile body **10** calculates a traveling path toward the target point **90** so as to avoid the surrounding obstacles on the basis of an environment map. At this time, a path searching algorithm is utilized, and when a target position is too far, the processing burden increases. Thus, the target positions **80** supplied to the mobile body **10** are set at intervals of about 10 m, for example, as shown in FIGS. 3A and 3B. In a case where an environment map is utilized for localization, information about the target positions **80** supplied to this mobile body **10** is plotted on the environment map. Here, the environment map on which the target positions **80** have been plotted shall be referred to as an advance map.

(81) The advance map is obtained by adding waypoint information (information about the target positions **80**) to the environment map. In an embodiment, a parameter for localization in accordance with the traveling environment is further embedded in advance in the advance map. Thus, the parameter for localization can be acquired from the advance map at a point corresponding to each of the target positions **80**.

(82) The mobile body **10** reads, together with information about the target position **80**, a parameter for localization to be utilized when moving to that target position **80** from the advance map, and performs localization processing utilizing the parameter suited to the traveling environment. This parameter includes the propriety of used sensors, a covariance value of the extended Kalman filter **300**, and settings necessary for each of the localization techniques such as the SLAM **130** described above.

(83) FIG. 4 is a schematic view showing a configuration of a system **2000** in a case of embedding environmental information in an advance map. As shown in FIG. 4, the system **2000** has an

advance map **400**, a parameter selection unit **500** (parameter changing unit), a localization unit **600**, a path planning unit **700**, and a vehicle control unit **800**. In this example, the advance map **400** is prepared in advance, and the advance map **400** includes an environment map **410**, a localization parameter **420**, and waypoint information **430**.

(84) The localization unit **600** receives a parameter to be utilized for localization, and performs localization processing utilizing that parameter. The localization unit **600** corresponds to the system **1000** shown in FIG. **1**. The parameter selection unit **500** receives destination information about which of the target positions **80** the path planning unit **700** is calculating a path for, selects the localization parameter **420** associated with that destination information, and passes the parameter to the localization unit **600**.

(85) The path planning unit **700** is equivalent to what is called a navigation device, and calculates a path along which the mobile body **10** intends to move. The path planning unit **700** calculates the path on the basis of the target position **80** and the position of the mobile body **10**, and sends the calculated path to the vehicle control unit **800**. In addition, as described above, the path planning unit **700** sends the target position **80** to the parameter selection unit **500**. The vehicle control unit **800** controls the mobile body **10** so as to follow the path.

(86) FIG. **5** is a flowchart showing processing in the case of embedding environmental information in the advance map. First, in step **S10**, information (target position) about the target point **90** is acquired. Accordingly, the target positions **80** are sequentially defined on the traveling path of the mobile body **10**. In next step **S12**, a parameter to be utilized for localization, selected by the parameter selection unit **500** on the basis of the target position **80**, is acquired from the advance map in which the environmental information has been embedded. In next step **S14**, localization computation is performed on the basis of the parameter acquired in step **S12**. In next step **S16**, the estimated position of the mobile body is output, and in next step **S18**, path planning processing is performed on the basis of the position of the mobile body. In next step **S20**, it is determined whether or not traveling to the target point **90** has been completed, and in a case where traveling has been completed, the process is terminated. On the other hand, in a case where traveling has not been completed, return is made to step **S10**.

(87) 5.2. Case in which Mobile Body Itself Recognizes Environment

(88) Next, a case in which the mobile body **10** itself recognizes the environment will be described. The mobile body **10** itself can recognize the environment with identification devices mounted. Examples of an identifier include an identification device that monitors a road surface condition, an identification device that monitors whether or not the sky is hidden by buildings, and an identification device that monitors whether there is no obstacle of a material that transmits light.

(89) The mobile body **10** selects a profile in which parameters for localization suited to the environment have been set, in accordance with a combination of identified information. For example, by indicating an identification result obtained by each of the identification devices by two values of "0" and "1", and setting a profile in accordance with their combination in advance, a profile in accordance with an environment identification result can be selected. The profile is a list of parameters for localization, and a parameter list of representative environments is generated in advance, and is supplied to a database of the mobile body **10**. The mobile body **10** determines environment identification information, and selects an appropriate profile included in the database to dynamically change the parameter for localization.

(90) FIG. **6** is a schematic view showing a configuration of a system **3000** in a case where a mobile body itself recognizes the environment. As shown in FIG. **6**, the system **3000** has an identification device **650**, an environment recognition unit **900**, a profile selection unit (parameter changing unit) **950**, a localization profile **980**, the advance map **400**, the localization unit **600**, the path planning unit **700**, and the vehicle control unit **800**. The advance map **400** includes the environment map **410** and the waypoint information **430**.

(91) The environment recognition unit **900** determines in what kind of traveling environment the

mobile body **10** is currently traveling on the basis of identification results obtained by a plurality of identification devices **650**, and makes a classification. Here, classes are obtained by classifying representative environments such as an environment in which it is difficult to use the GPS **140** and an environment in which it is difficult to use the wheel odometry **110**.

(92) The identification device **650** mainly receives an image as an input, and determines by machine learning or the like in advance whether a feature requested to be extracted appears in an image, and makes a notification to the environment recognition unit **900** that performs processing of determining a class. Machine learning herein assumes, for example, learning in which an image is input to determine whether a feature is included, such as deep learning or a method of matching a template image. The environment recognition unit **900** integrates identification results obtained by the large number of identification devices **650**, and makes a classification about which profile the environment utilizes. Deep learning can also be used for the classification.

(93) The profile selection unit **950** selects a localization profile **980** suited to a classified environment, and sends the localization profile **980** to the localization unit **600**. The localization unit **600** performs localization processing using parameters of the localization profile **980**. Thereafter, the path planning unit **700** generates a path, and the vehicle control unit **800** controls the mobile body **10**.

(94) FIG. 7 is a schematic view showing an example of the localization profile **980**. As shown in FIG. 7, in the localization profile **980**, localization-related parameters such as covariance values of the SLAM **130**, the GPS **140**, and the like are defined in advance in accordance with the environment determined by the environment recognition unit **900**. Note that the example of FIG. 7 shows the square **30** and the glass walls **52** as an example of a profile in accordance with the traveling environment. In accordance with identification results obtained by the plurality of identification devices described above, it is determined whether the traveling environment is the square **30** or the glass walls **52** shown in FIG. 7. In a case where it is determined that the traveling environment is the square **30**, the profile for “square” shown in FIG. 7 is selected. In this profile for “square”, covariance values of the SLAM **130**, the GPS **140**, and the like, as well as parameters for various sensors for localization are set in advance. Consequently, by selecting a profile in accordance with the traveling environment, a parameter for localization can be acquired.

(95) FIG. 8 is a flowchart showing processing in a case where the mobile body itself recognizes an environment. First, in step **S30**, the environment recognition unit **900** identifies the environment. In next step **S32**, the profile selection unit **950** selects the localization profile **980** in accordance with an environment determination result. In next step **S34**, computation for localization is performed on the basis of the localization profile **980** selected in step **S32**. In next step **S36**, the position of the mobile body is output on the basis of a result of computation in step **S34**. In next step **S38**, path planning processing is performed on the basis of the position of the mobile body. In next step **S40**, it is determined whether or not traveling to the target point **90** has been completed, and in a case where traveling has been completed, the process is terminated. On the other hand, in a case where traveling has not been completed, return is made to step **S30**.

(96) 5.3. Example of Embedding Environmental Information in Advance Map and Performing Path Planning in Accordance with Sensor Situation

(97) By embedding parameters for localization in the environment map, an optimum path can be selected from the current sensor situation of the mobile body **10**. For example, in a case of going to a certain destination, assume that there are two alternatives of an indoor path which is the shortest path but is surrounded by the glass walls **52** and a path which is indirect but is outdoors with the open sky. In this case, when the former path is selected, the mobile body **10** will travel mainly using the wheel odometry **110** and the IMU **100**. On the other hand, when the latter path is selected, the mobile body **10** will travel mainly using the GPS **140**, the wheel odometry **110**, and the IMU **100**.

(98) In a case where all sensors can be utilized without any problem, it is desirable to travel along

the former path as the shortest path. On the other hand, in a case where the wheel odometry **110** has been failed, it is predicted that localization cannot be performed with the former path.

(99) Thus, in a case where a failure of the wheel odometry **110** is considered, it is desirable to travel along the latter path. By embedding parameters for localization in the advance map, it is also possible to dynamically switch between the traveling paths while referring to a sensor situation of the mobile body **10**. That is, by embedding parameters in the advance map in advance, a path on which the position of the mobile body is less likely to be lost can be selected even in a case where there is a failed sensor.

(100) FIG. **9** is a schematic view showing a configuration of a system **4000** in a case of performing path planning in accordance with a map having parameters embedded and a sensor situation. As shown in FIG. **9**, the system **4000** has a sensor monitoring unit **640** in addition to the configuration of FIG. **4**. FIG. **9** depicts various sensors such as the camera **200**, the laser scanner **210**, a wheel encoder **112**, the IMU **100**, the GPS **140**, and the magnetic sensor **150** in addition to the sensor monitoring unit **640**. Note that the wheel encoder **112** is used in the wheel odometry **110**. The sensor monitoring unit **640** monitors the various sensors such as the camera **200**, the laser scanner **210**, the wheel encoder **112**, the IMU **100**, the GPS **140**, and the magnetic sensor **150**. When it is determined that these sensors have been failed, the sensor monitoring unit **640** makes a notification to the path planning unit **700**.

(101) For example, if any sensor continues outputting outliers, the sensor monitoring unit **640** notifies the path planning unit **700** of a failure flag of that sensor. The path planning unit **700** refers to the parameters embedded in the advance map to select such a target position that the position of the mobile body is not lost with a configuration excluding the failed sensor. In a case where the covariance of a parameter embedded in the map is small for the sensor (failed sensor) for which a failure flag has been set up, the position of the mobile body is highly likely to be lost, so that another target position is selected, and path planning calculation is performed again. In this manner, a path on which the position of the mobile body is less likely to be lost can be selected in accordance with the sensor situation by using the map having parameters embedded.

(102) FIG. **10** is a flowchart showing processing in a case of performing path planning in accordance with the map having parameters embedded and a sensor situation. First, in step **S50**, information (target position) about the target point **90** is acquired. Accordingly, the target positions **80** are sequentially defined on the traveling path of the mobile body **10**. In next step **S52**, a parameter to be utilized for localization, selected by the parameter selection unit **500** on the basis of the target positions **80**, is acquired from the advance map in which the environmental information has been embedded. In next step **S54**, the sensor monitoring unit **640** acquires sensor failure information. In next step **S56**, it is determined whether or not the sensor is normal on the basis of the sensor failure information, and in a case where the sensor is normal, the process proceeds into step **S58**. In step **S58**, computation for localization is performed. On the other hand, in a case where the sensor is abnormal, the process proceeds into step **S57**, and after changing the target position, return is made to step **S52**.

(103) The process proceeds into step **S60** after step **S58**, and the estimated position of the mobile body is output. In next step **S62**, path planning processing is performed on the basis of the position of the mobile body. In next step **S64**, it is determined whether or not traveling to the target point **90** has been completed, and in a case where traveling has been completed, the process is terminated. On the other hand, in a case where traveling has not been completed, return is made to step **S50**.

(104) The present disclosure can be applied to various platforms in addition to a wheel-type mobile body. As long as an autonomous mobile body such as an autonomous car, a personal mobility, a humanoid, or a drone senses to perform localization, the application range is not particularly limited. In a mobile body to which the present disclosure has been applied, a phenomenon in which a localization result includes an error under a specific environment and a phenomenon in which an autonomous movement can no longer be performed accordingly are significantly improved. Since a

parameter specially designed for an environment can be changed dynamically, the localization accuracy is improved and robustness can be improved in each environment.

(105) As an effect obtained by improvement in localization accuracy and robustness, an effect of improvement in path followability in which the localization accuracy with respect to a path of the mobile body is improved, so that finer vehicle control can be made is obtained. In addition, in a flying platform such as a drone, the stopping accuracy at the time of hovering is improved, and improvement in stopping accuracy, such as stabilization of flying shooting, can be achieved. Further, with the improvement in robustness, environments that can be adapted to in a single travel increase, so that a range that can be traveled in a single travel is enlarged, and enlargement of the range that can be traveled can be achieved.

(106) The localization technology is an important factor that improves basic performance of a mobile body. The present disclosure significantly contributes to the application development through use of a mobile body which will be further accelerated in the future by improving basic performance.

(107) Embodiment(s) of the present disclosure have been described above with reference to the accompanying drawings, however the present disclosure is not limited to the above examples. A person skilled in the art may find various alterations and modifications within the scope of the appended claims, and it should be understood that they will naturally come under the technical scope of the present disclosure.

(108) Further, the effects described in this specification are merely illustrative or exemplified effects, and are not limitative. That is, with or in the place of the above effects, the technology according to the present disclosure may achieve other effects that are clear to those skilled in the art from the description of this specification.

(109) Additionally, the present disclosure may also be configured as below.

(110) (1) A mobile body including:

(111) a localization unit configured to estimate a position of the mobile body on the basis of a parameter for localization; and

(112) a parameter changing unit configured to dynamically change the parameter in accordance with a traveling environment in which the mobile body is traveling.

(113) (2) The mobile body according to (1), in which the parameter is a parameter corresponding to a sensor used for the localization, a value of covariance in an extended Kalman filter that fuses a plurality of schemes for localization, or a set value in each of the schemes.

(114) (3) The mobile body according to (1) or (2), in which the parameter is a parameter corresponding to IMU, wheel odometry, visual odometry, SLAM, or GPS used for the localization.

(115) (4) The mobile body according to any one of (1) to (3), in which the parameter is defined in advance in correspondence with waypoint information in an environment map, and

(116) the parameter changing unit applies a waypoint of a traveling path to the waypoint information to change the parameter.

(117) (5) The mobile body according to any one of (1) to (4), further including: an environment recognition unit configured to recognize the traveling environment at a time of traveling, in which (118) the parameter changing unit changes the parameter in accordance with the traveling environment recognized at the time of traveling.

(119) (6) The mobile body according to any one of (1) to (5), in which a profile of the parameter in accordance with the traveling environment is registered in advance, and

(120) the parameter changing unit changes the parameter on the basis of the profile obtained from the traveling environment recognized at the time of traveling.

(121) (7) The mobile body according to any one of (1) to (6), in which the parameter is defined in advance in correspondence with waypoint information in an environment map, and

(122) the parameter changing unit applies a waypoint of a traveling path to the waypoint information to change the parameter,

(123) the mobile body further including: a sensor monitoring unit configured to monitor a state of a sensor used for the localization; and

(124) a path selection unit configured to select a path on the basis of the parameter in a different traveling path and a state of the sensor corresponding to the parameter.

(125) (8) A method of controlling a mobile body, including:

(126) estimating a position of the mobile body on the basis of a parameter for localization; and

(127) dynamically changing the parameter in accordance with a traveling environment in which the mobile body is traveling.

(128) (9) A program for causing a computer to function as:

(129) means for estimating a position of the mobile body on the basis of a parameter for localization; and

(130) means for dynamically changing the parameter in accordance with a traveling environment in which the mobile body is traveling.

(131) (10) A mobile body controller including:

(132) circuitry configured to recognize an environment surrounding a mobile body to be controlled, and change parameters used for self-position estimation by the mobile body based on the recognized environment.

(133) (11) The mobile body controller according to (10), wherein the circuitry recognizes the environment surrounding the mobile body by recognizing a traveling environment in which the mobile body travels.

(134) (12) The mobile body controller according to (10) or (11), wherein the traveling environment includes a planned path from a current position of the mobile body to a target position of the mobile body.

(135) (13) The mobile body controller according to any one of (10) to (12), wherein the circuitry recognizes the environment surrounding the mobile body by recognizing at least one of a ground surface condition, a presence of buildings around the mobile body, or a presence of materials that transmit light around the mobile body.

(136) (14) The mobile body controller according to any one of (10) to (13), wherein the circuitry recognizes the environment surrounding the mobile body by recognizing whether wheels of the mobile body are likely to slip.

(137) (15) The mobile body controller according to any one of (10) to (14), wherein the changed parameters used for self-position estimation relate to at least one of wheel odometry, simultaneous localization and mapping (SLAM), map matching, a laser scanner, or a global positioning system (GPS) used to estimate a position of the mobile body.

(138) (16) The mobile body controller according to any one of (10) to (15), wherein the changed parameters used for self-position estimation relate to a localization profile.

(139) (17) The mobile body controller according to any one of (10) to (16), wherein the environment surrounding the mobile body is recognized using one or more sensors included in the mobile body.

(140) (18) A mobile body including:

(141) one or more sensors; and

(142) a mobile body controller, the mobile body controller including circuitry configured to recognize an environment surrounding the mobile body controlled by the mobile body controller, and change parameters used for self-position estimation by the mobile body based on the recognized environment.

(143) (19) A mobile body control method including:

(144) recognizing an environment surrounding a mobile body to be controlled; and

(145) changing parameters used for self-position estimation by the mobile body based on the recognized environment.

(146) It should be understood by those skilled in the art that various modifications, combinations,

sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

REFERENCE SIGNS LIST

(147) **500** Parameter selection unit **600** Localization unit

Claims

1. A mobile body controller comprising: circuitry configured to recognize an environment surrounding a mobile body to be controlled, and change one or more parameters to determine a weight of each sensor used for self-position estimation by the mobile body based on a determined class at each point along a planned path of the mobile body through the recognized environment, wherein the class is determined based on an input image used to recognize the environment, wherein the determined class of the recognized environment is determined based on a degree of slip associated with a recognized ground surface condition at each point along the planned path of the mobile body, and wherein the circuitry changes at least one parameter to determine a weight of wheel odometry used for the self-position estimation according to the degree of slip associated with the recognized ground surface condition at each point along the planned path of the mobile body.
2. The mobile body controller according to claim 1, wherein the circuitry recognizes the environment surrounding the mobile body by recognizing a traveling environment in which the mobile body travels.
3. The mobile body controller according to claim 2, wherein the traveling environment includes the planned path from a current position of the mobile body to a target position of the mobile body.
4. The mobile body controller according to claim 1, wherein the determined class of the recognized environment is further determined by recognizing at least one of a presence of buildings around the mobile body or a presence of materials that transmit, reflect, or diffract light around the mobile body.
5. The mobile body controller according to claim 1, wherein the circuitry recognizes the environment surrounding the mobile body by recognizing whether wheels of the mobile body are likely to slip.
6. The mobile body controller according to claim 1, wherein the one or more changed parameters used for self-position estimation further relate to at least one of simultaneous localization and mapping (SLAM), map matching, a laser scanner, or a global positioning system (GPS) used to estimate a position of the mobile body.
7. The mobile body controller according to claim 1, wherein the one or more changed parameters used for self-position estimation relate to a localization profile.
8. The mobile body controller according to claim 1, wherein the environment surrounding the mobile body is recognized using one or more sensors included in the mobile body.
9. A mobile body comprising: one or more sensors; and a mobile body controller, the mobile body controller including circuitry configured to recognize an environment surrounding the mobile body controlled by the mobile body controller, and change one or more parameters to determine a weight of each sensor used for self-position estimation by the mobile body based on a determined class at each point along a planned path of the mobile body through the recognized environment, wherein the class is determined based on an input image used to recognize the environment, wherein the determined class of the recognized environment is determined based on a degree of slip associated with a recognized ground surface condition at each point along the planned path of the mobile body, and wherein at least one parameter is changed to determine a weight of wheel odometry used for the self-position estimation according to the degree of slip associated with the recognized ground surface condition at each point along the planned path of the mobile body.
10. A mobile body control method comprising: recognizing an environment surrounding a mobile body to be controlled; and changing one or more parameters to determine a weight of each sensor

used for self-position estimation by the mobile body based on a determined class at each point along a planned path of the mobile body through the recognized environment, wherein the class is determined based on an input image used to recognize the environment, wherein the determined class of the recognized environment is determined based on a degree of slip associated with a recognized ground surface condition at each point along the planned path of the mobile body, and wherein at least one parameter is changed to determine a weight of wheel odometry used for the self-position estimation according to the degree of slip associated with the recognized ground surface condition at each point along the planned path of the mobile body.

11. The mobile body controller according to claim 1, wherein the circuitry is further configured to acquire the input image.

12. The mobile body controller according to claim 1, wherein the circuitry changes the at least one parameter to determine the weight of the wheel odometry used for the self-position estimation such that a covariance of the wheel odometry with the self-position estimation is smaller as the degree of slip decreases.
